

Sameer Asif

Islamabad, Pakistan | +92 323 5151542 | sameerasif189@gmail.com | [GitHub](#) | [LinkedIn](#)

SUMMARY

AI Engineer and final-year BS Artificial Intelligence student at FAST-NUCES, building production LLM and RAG systems at Infigo Solutions. Experienced across the full lifecycle: data scraping and pipelines, model training (GANs, U-Nets, Transformers) in PyTorch, ONNX deployment, FastAPI services, and Postgres/SQL Server databases.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C/C++, JavaScript/TypeScript, Bash
- **AI / ML:** PyTorch, Hugging Face Transformers, GANs, U-Net, EfficientNet, multi-task learning, self-attention, RAG, pgvector, LoRA/SFT, prompt engineering
- **CV / Speech / NLP:** OpenCV, timm, multi-scale detection, facial-expression recognition, faster-whisper, Spark-TTS, librosa, N-gram language models
- **Data / Backend:** FastAPI, WebSockets, async Python, Playwright, SQL Server (pyodbc), Postgres (Neon) + pgvector, SQLAlchemy, pandas, NumPy
- **Deployment / Tools:** ONNX Runtime, Unity Barracuda/Sentis, CUDA 12, Vercel, Docker, Git, Linux/WSL, React, Vite

EXPERIENCE

Junior AI Engineer | Infigo Solutions, Islamabad | July 2025 – Present

- Lead engineer for the customer-support agent platform — a multi-channel (web, WhatsApp, Slack) FastAPI app on Neon Postgres + pgvector, deployed on Vercel and live on infigosolutions.com.
- Built a hybrid RAG pipeline (chunking, embeddings, cosine + Postgres FTS search, ERP context joins) with cost/safety guardrails on Groq Llama-3.1-8B: answer caching, budget tracker, and retrieval/output caps.
- Shipped an embeddable React chat widget and public chat API; authored the Postgres migration suite, synthetic ERP seeder, and smoke-test harness.

AI Engineering Intern | Infigo Solutions, Islamabad | June 2025 – July 2025

- Built the first Infigobot prototype (FastAPI + Groq LLM + curated KB), the Vercel deployment, and the site-key auth flow used in production; promoted to full-time based on this work.

PROJECTS

Adaptive Psychological Horror Game — Final Year Project | [GitHub](#)

- Trained facial-expression recognition across 7 datasets (DFEW, FER2013, RAF-DB, CK+, Aff-Wild2, etc.) mapped to a 4-class stress schema feeding a unified video + audio + heart-rate classifier consumed by a Unity game.
- Built a stress-conditioned Fusion Generator (~500K-param MLP), a conditional WaveGAN for horror audio, and a U-Net for PBR texture corruption — all exported to ONNX for Unity Barracuda/Sentis. *(PyTorch, ONNX, OpenCV, Unity)*

Lead-Generation Scraping Engine & SQL Data Bank | [GitHub](#)

- Built an async Playwright scraping suite (Clutch, Wellfound) with custom anti-bot stealth patches to extract company profiles, reviews, and contact details at scale, with regex email/phone/LinkedIn enrichment.
- Designed a normalized SQL Server schema (Source → Organizations → Contacts → Reviews) with deduplication, foreign-key reconciliation, and incremental sync. *(Python, Playwright, SQL Server/pyodbc, pandas)*

Local Voice-Cloning Conversational Agent | [GitHub](#)

- End-to-end voice pipeline (faster-whisper STT → Groq Llama-3.1-8B → Spark-TTS cloned voice) in 16 GB VRAM; FastAPI + WebSockets backend and React/Vite SPA with consent-gated voice upload. *(FastAPI, PyTorch, Spark-TTS, React)*

Hybrid Wildlife Detection | [GitHub](#)

- 4.5M-param detector synthesizing 3 research papers: EfficientNet-B0 multi-scale fusion + spatial attention + 3 task heads (classification, bbox, objectness), trained on Missouri Camera Traps tuned for a 6 GB GPU. *(PyTorch, timm, alumentations)*

From-Scratch Transformer + RAG (NLP) | [GitHub](#)

- Built a multi-task Transformer encoder/decoder from scratch (82% sentiment accuracy) with retrieval-augmented explanation generation and perplexity analysis. *(PyTorch, NumPy)*

Additional: [BBC Urdu N-gram text generator](#) (Streamlit); [Sentinel adaptive trading bot](#) (Binance API, sentiment signals); [SIP investment calculator](#) (React/Vite).

EDUCATION

BS Artificial Intelligence | FAST-NUCES, Islamabad | 2022 – 2026 (Expected) Coursework: Deep Learning, NLP, Computer Vision, Generative AI, Databases, Software Engineering, DSA, Probability & Statistics

Cambridge A Levels | Beaconhouse School System | 2020 – 2022 | **Cambridge O Levels** | Beaconhouse | 2017 – 2020